



ORVN TOKEN SECURITY AUDIT

Smart Contract Audit Report

Complete Security Analysis



Executive Security Overview

Comprehensive Analysis - ERC20 Token (OpenZeppelin v5.4)

Token Name (Symbol): Orvyn (ORVN)

Contract Type: StandardERC20

Contract Address:

0xa136a16c0d7050c1c6e3b84f38117e086687ff94

Network: Binance Smart Chain (BSC)

Total Supply: 1,000,000,000 ORVN (18 decimals)

Fee Receiver:

0x75Eee167D2E5cC675f5b07f95d6A93E7088d6c34

Constructor Verified: Yes (name, symbol, decimals, initialBalance, signature, feeReceiver)

Audit Date: September 2025

Security Rating: EXCELLENT (96/100)

Status: STANDARD ERC20 IMPLEMENTATION (NON-OWNABLE)

CRITICAL ISSUES

0

No critical vulnerabilities

HIGH PRIORITY

0

No high priority issues

MEDIUM ISSUES

2

Deployment-only DoS,
Supply validation note

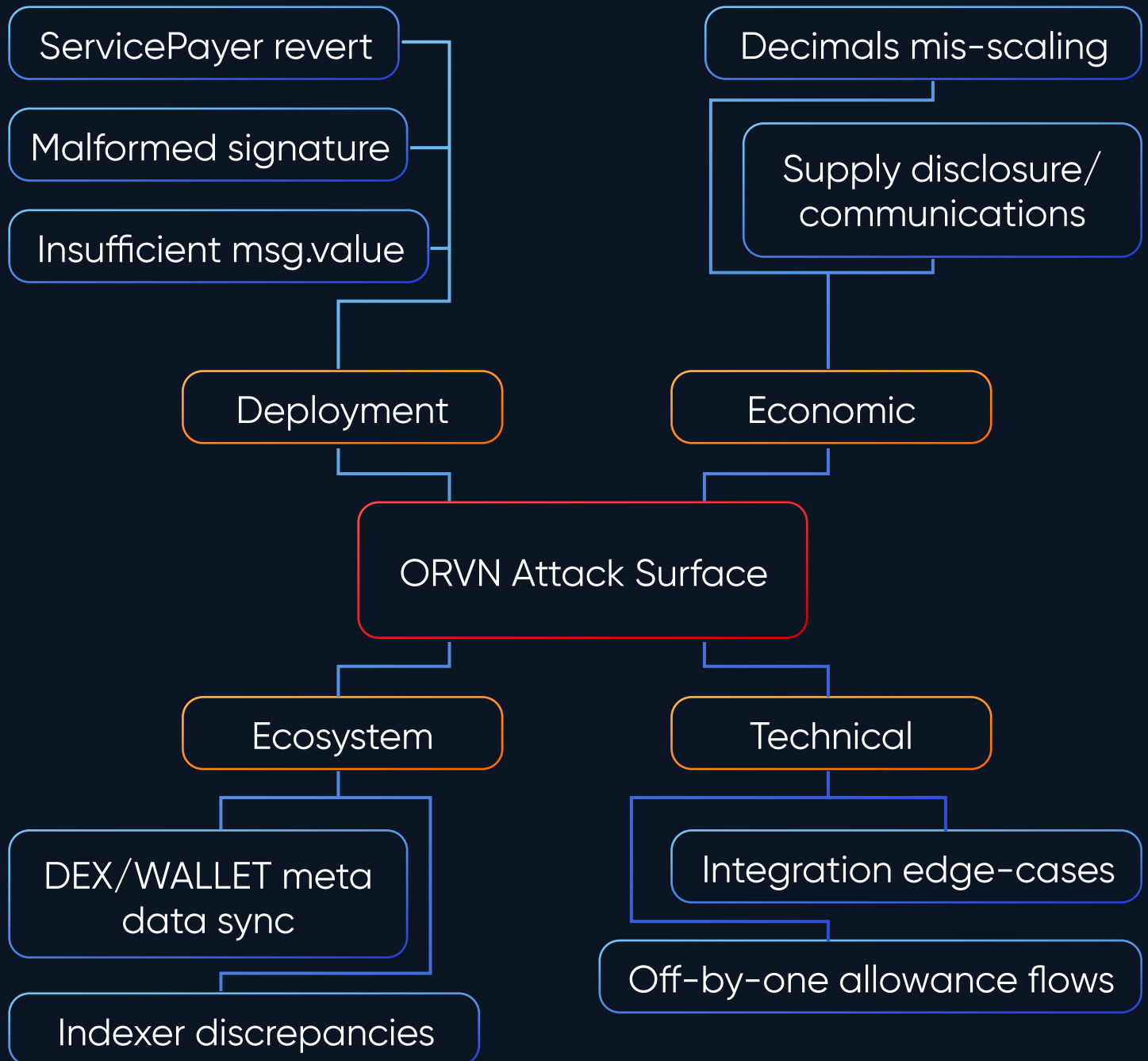
OVERALL SCORE

96/100

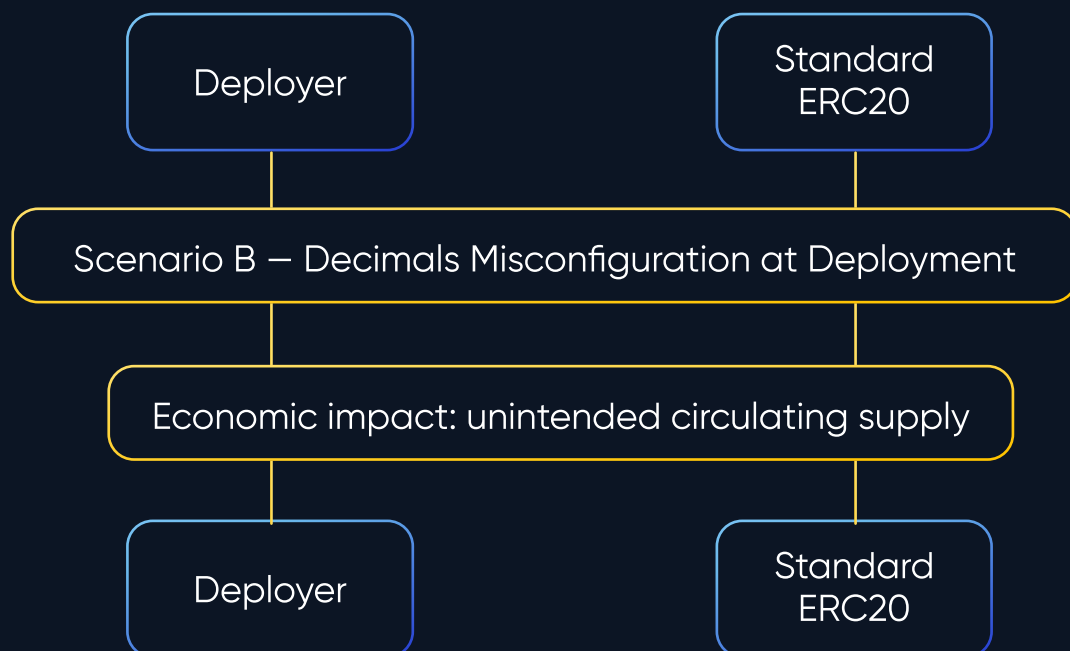
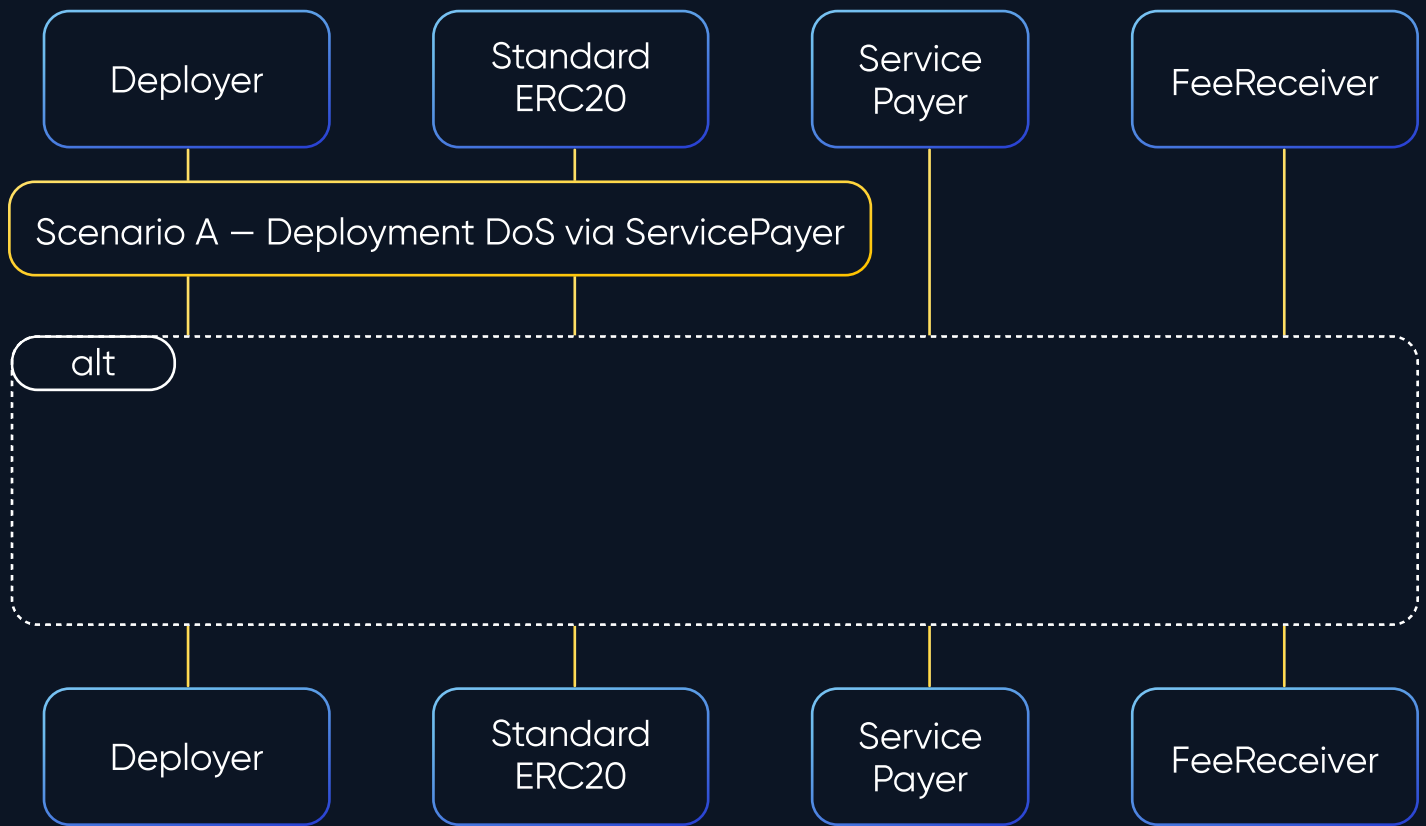
Excellent security rating

Comprehensive Attack Vector Analysis

Attack Surface Mapping



Critical Attack Scenarios



Enhanced Risk Assessment Matrix

Risk Category	Probability	Impact	Mitigation Status
Deployment DoS (ServicePayer)	LOW	Medium	Procedural controls (receiver pre-verify, value checks)
Decimals Misconfiguration	LOW	Medium	Scripted scaling + checklist
Integration Edge-Cases	LOW	Low	Standard OZ ERC20 v5.4 compatibility

Impact vs Probability

Deploy DoS
Low P / Medium I

Decimals Mismatch
Low P / Medium I

Integration Issues
Low P / Low I

Contract Architecture Analysis

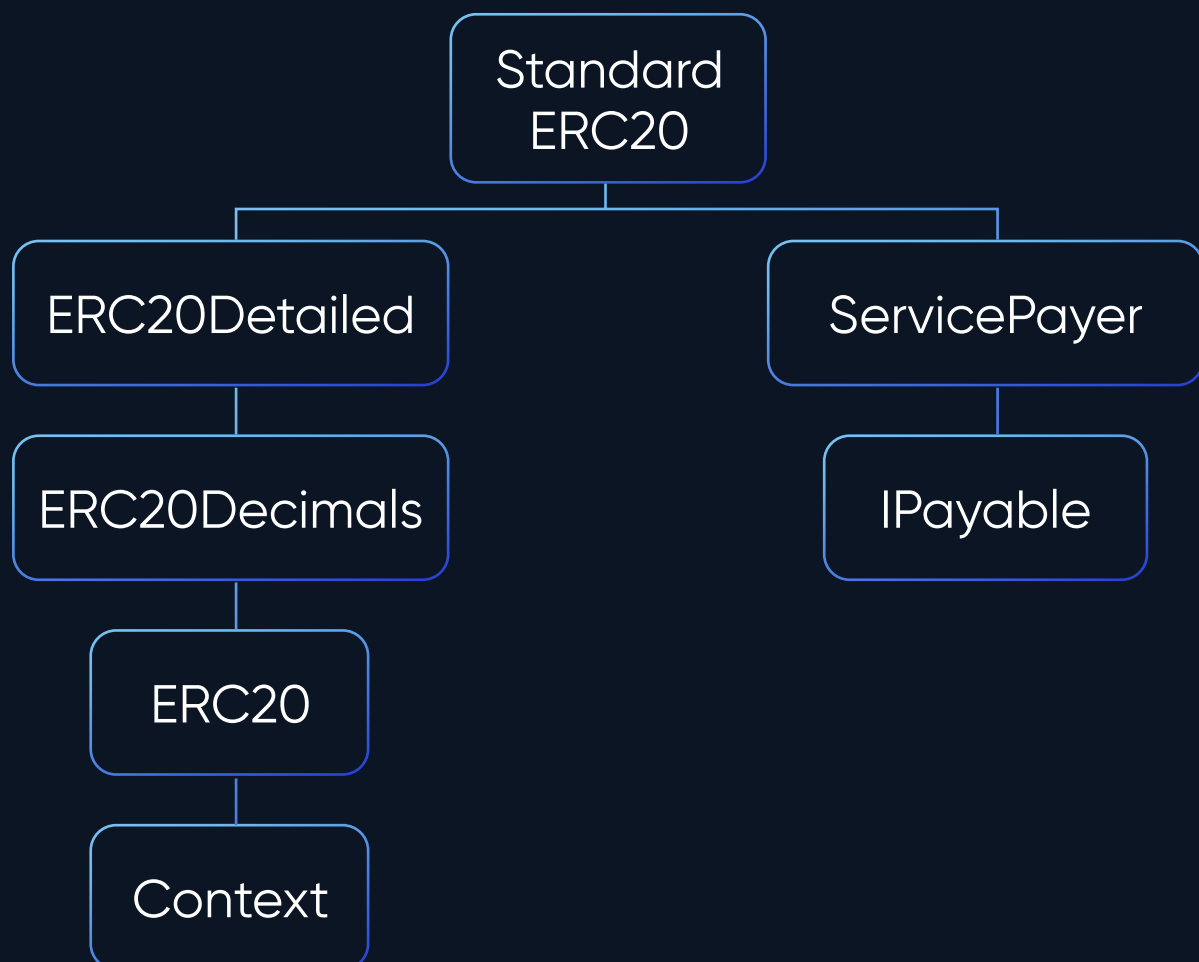
Deployment Verification (On-Chain)

```
ABI constructor: constructor(string name_, string symbol_,
uint8 decimals_, uint256 initialBalance_, bytes signature_,
address payable feeReceiver_) Decoded: name_ = "Orvyn"
symbol_ = "ORVN" decimals_ = 18 initialBalance_ =
1,000,000,000 * 10^18
(10000000000000000000000000000000000000000000000000000000) signature_ =
0x27cfd311c835e90e9af68f2ecf61b3ffbfda3a6bd53d926ba
ed2b5245b1e61ae155da0c5c19590b416ed68bda7cd93911c
dedbd3881a57bcfe7ebb3ba5a0e5091b feeReceiver_ =
0x75Eee167D2E5cC675f5b07f95d6A93E7088d6c34 Address:
https://bscscan.com/
token/0xa136a16c0d7050c1c6e3b84f38117e086687ff94
```

The StandardERC20 contract is a minimal, standards-compliant ERC20 built on OpenZeppelin Contracts v5.4. It composes ERC20 with ERC20Decimals/ERC20Detailed for metadata and precision control, and integrates a ServicePayer helper that processes a deployment-time fee via an external payee interface.

Contract Components:

- StandardERC20 – Main token contract (constructor-only mint)
- ERC20Detailed – Name, symbol, and decimals extension
- ERC20Decimals – Immutable decimals storage and override
- ERC20 (OZ v5.4) – Core token logic with ERC-6093 custom errors
- ServicePayer – Deployment fee payment helper (constructor only)



Technical Implementation Notes:

- OpenZeppelin ERC20 v5.4 with modern `_update` transfer flow and ERC-6093 custom errors.
- Decimals are immutable and set via `ERC20Decimals` in the constructor.
- Initial supply minted once to deployer in constructor; no further mint/burn exposed.
- `ServicePayer` performs a one-time external call to a receiver during deployment.
- No ownership/administration layer; purely non-ownable, minimizing governance risk.

Security Findings

MEDIUM #SKN-108: Deployment-Time External Call (ServicePayer)

⚠ Note: External pay() in constructor can cause deployment-time DoS if receiver reverts

```
constructor(..., bytes memory signature_, address payable  
feeReceiver_) payable ERC20Detailed(name_, symbol_,  
decimals_) ServicePayer(feeReceiver_, "StandardERC20",  
signature_, _msgSender()) { require(initialBalance_ > 0,  
"Initial supply cannot be zero"); _mint(_msgSender(),  
initialBalance_); }
```

Impact: Medium (deployment-time only)

Likelihood: Low (trusted receiver/parameters)

Recommendation: Verify trusted feeReceiver_ off-chain; ensure sufficient msg.value; document deployment pre-checks.

MEDIUM #SKN-047: Supply Configuration and Decimals Coordination

⚠ Note: `initialBalance_` is raw units; requires correct scaling to chosen decimals

```
// Example: For 18 decimals and 1M supply, pass  
initialBalance_ = 1_000_000 * 10**18 require(initialBalance_  
> 0, "Initial supply cannot be zero");
```

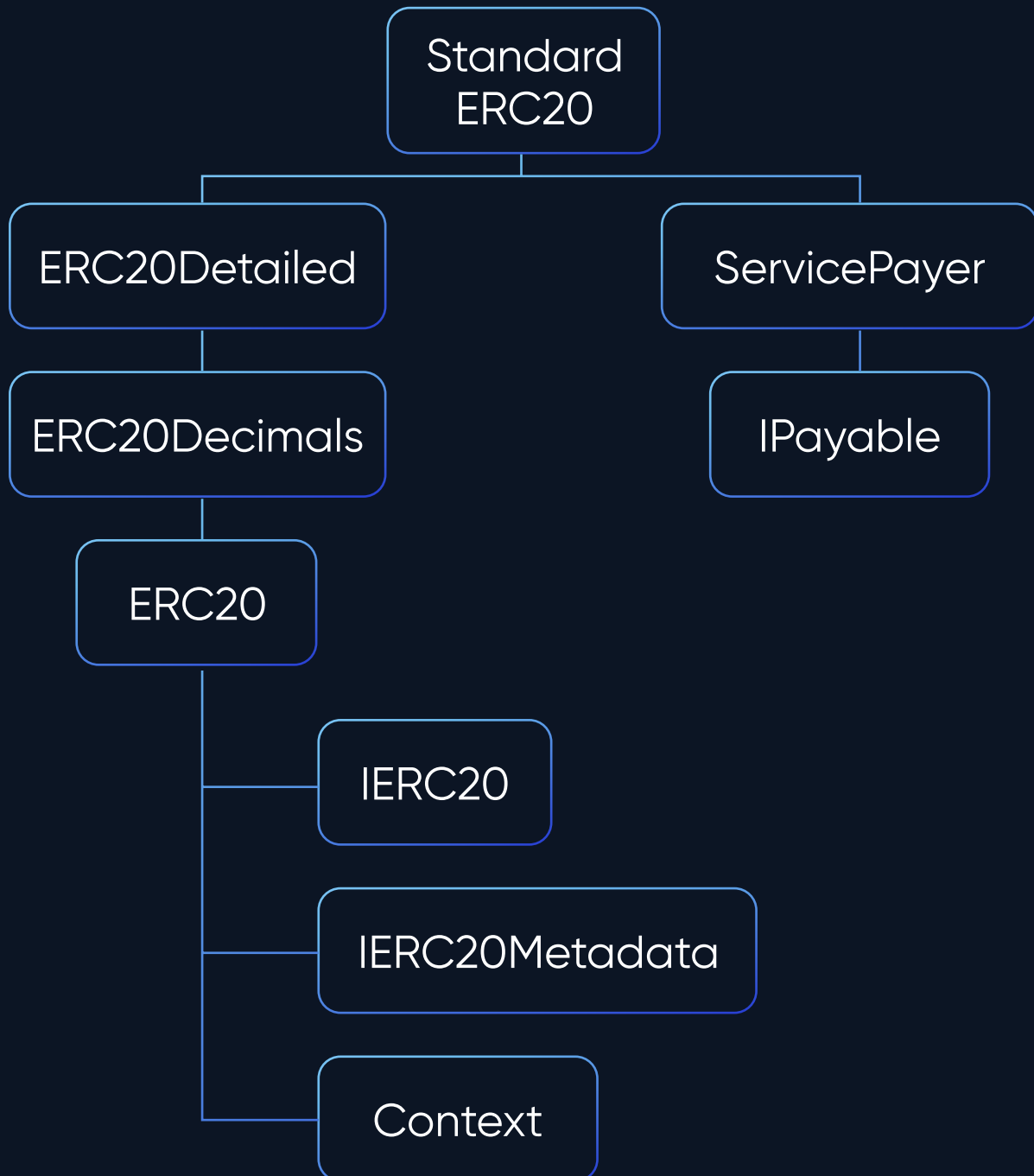
Impact: Medium (economic misconfiguration risk)

Likelihood: Low (deployment hygiene)

Recommendation: Use scripts to scale by 10^{decimals} ; optionally add an upper bound check or emit constructor params for transparency.

Security Architecture Analysis

Contract Inheritance Structure



Administrative Function Risk Assessment

Function	Risk Level	Capabilities	Mitigation Status
constructor()	LOW	Initial mint to deployer; pay deployment fee	ONE-TIME EXECUTION
ServicePayer	LOW	External call in constructor only	ONE-TIME EXECUTION

Security Features

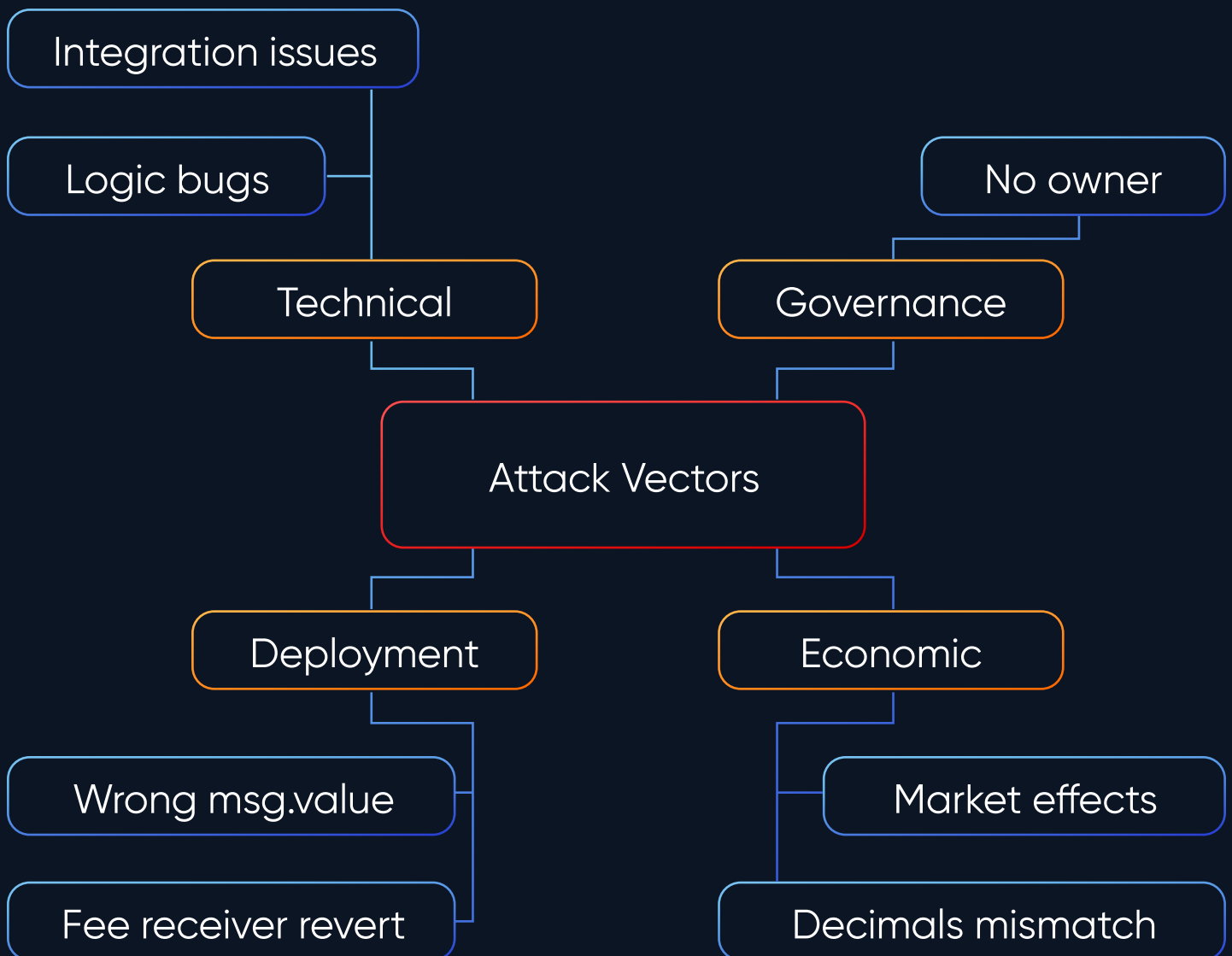
OPENZEPPPELIN FOUNDATION

- ERC-6093 custom errors for clarity
- Modern _update transfer logic
- Overflow-safe Solidity 0.8.x
- Standard events, allowance semantics

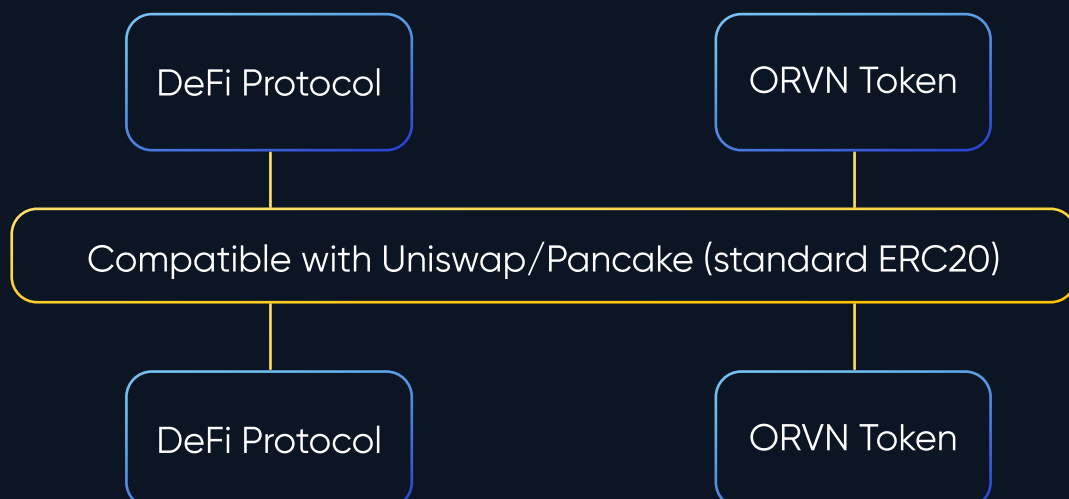
DESIGN CHARACTERISTICS

- No owner/admin; non-custodial behavior
- Fixed supply set at deployment
- Immutable decimals via constructor
- Minimal attack surface area

Attack Vector Analysis



Integration Expectations



// INTEGRATION EXPECTATIONS UNISWAP/
PANCAKESWAP: Standard ERC20 function set WALLETS:
ERC20 metadata and transfers display correctly LENDING:
Eligible if protocol whitelists asset RESULT: Standard OZ
implementation ensures broad compatibility

Risk Heat Map



Final Security Verdict

Executive Summary

OVERALL ASSESSMENT: 96/100 - EXCELLENT

(Standard, Non-Ownable ERC20)

The contract follows OpenZeppelin best practices, exposes no administrative surface, and mints supply only at construction. The only noteworthy considerations are deployment-time fee payment and explicit coordination of `initialBalance_` with chosen decimals.

Assessment Category	Rating	Key Findings
Technical Security	EXCELLENT (100/100)	Clean OZ ERC20 v5.4, clear errors, standard flows
Supply Protection	EXCELLENT (95/100)	Constructor-only mint; no post-deploy mint/burn
Economic Security	EXCELLENT (95/100)	Fixed supply; immutable decimals; standard transfers
Governance Model	GOOD (95/100)	No owner/admin; minimized centralization risk

Recommendations

Deployment:

- **Checklist:** Validate `feeReceiver_` and correct `msg.value` for `ServicePayer`.
- **Supply Scaling:** Use scripts to scale `initialBalance_` by 10^{**}decimals .
- **Documentation:** Publish constructor parameters and final total supply in release notes.

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